

Quantum Gravitation

Lecture 9

Renormalization and Parameter Matching

- Degrees of freedom
- Quantization
- Spectrum and response
- Consistency checks
- Example

■ Degrees of freedom

We separate propagating polarizations from constrained components before quantization.

$$g_i^0 = \mu^{d_i \varepsilon} (g_i + \delta g_i)$$

■ Quantization

Canonical normalization is fixed before creation and annihilation operators are introduced.

$$\beta_i = \mu dg_i/d\mu$$

■ Spectrum and response

Poles, residues, and group velocity are read from the same quadratic operator.

$$g_i(\mu_{obs}) = U_{ij} g_j(\mu_{calc})$$

■ Consistency checks

Hermiticity, positive norm, source conservation, and the low-energy limit are checked explicitly.

■ Example

A short numerical estimate fixes the scale of the effect before a full fit is attempted.

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Reading: Seo, ch. 9; RENORM-66 manual; RG Block Registry 2026